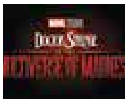
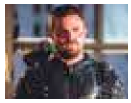


Horror-esque Doctor Strange
It has been confirmed that Doctor Strange: In the Multiverse of Madness will be a sort of horror film. **https://dgit.in/aug19-89**



Happy ending for Arrow
Stephen Amell confirmed that the series finale of Arrow will be "a super happy ending". **https://dgit.in/aug19-90**

Boo Science!

Boo man has enough of some aspects of the scientific community

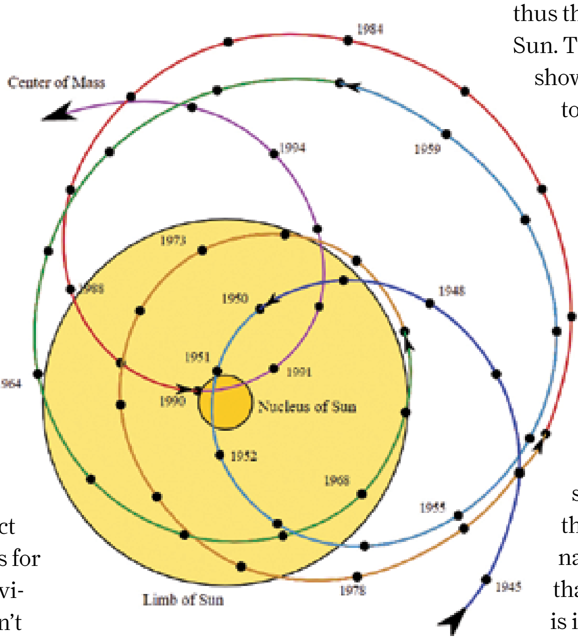
Boo-Man | boo@digit.in

Apparently I really do have the knack of finding fault with everything! However, this month's column is especially upsetting me since I've put my faith in science for so long now, that I feel especially let down.

In mid-July, as I was scrolling through my social media news feeds, I came across a controversial story. The story that was doing the rounds was that a study published in Scientific Reports (https://dgit.in/1908Boo1) had found that the Sun was causing global warming, and that the increases in warming could be explained by natural phenomenon. Of course, this is big news, because as far as we know, the recent rapid increase in climate change is caused by humans. It's supposed to be settled science, with more direct evidence for it available than there is for General Relativity or Evolution! Obviously, anything that suggests this isn't the case will, and should be scrutinised extremely carefully.

Normally, Scientific Reports is a very reputed journal, with pretty decent peer review legacy. It is published by Nature, after all, and claims to be "An open-access, multidisciplinary journal from Nature Research dedicated to constructive, inclusive, and rigorous peer review." One would expect that a controversial paper like this one, especially since it comes from an author who has been criticised in the past as well – Valentina Zharkova – would be poured over with great care. The paper looks at oscillations in the solar magnetic field and solar irradiance, it makes some very basic errors in orbital physics.

For example, the paper is being criticised by astrophysicists for assuming that the distance between the Sun and Earth changes based on the barycenter of the Sun-Jupiter system. A barycenter is the centre of mass of two bodies. If both bodies have equal mass, the barycenter lies halfway between them. If one of the bodies is much bigger than the other, the centre of mass lies closer to the body with larger mass, and could even be located inside the larger body. For example, the centre of mass of the Earth-Moon system lies at a point inside the Earth about 1,707 km below the surface. The centre of the Moon and the centre of the Earth orbit that point of space.



Barycenters are complex, because although we usually measure them as two-body problems, in practice the Sun, all of the planets, their moons, all of the asteroids, comets, bodies in orbit beyond Neptune, all of the stuff in the Oort cloud, and maybe more, all affect one another gravitationally, and it's impossible to plot all of this accurately. One could, in theory, painstakingly do the math for every single bit of space dust, but we tend to use approximations that ignore tiny bodies (in comparison) like the Earth and the inner planets. Jupiter and Saturn, by far, cause the shifts in SSB (solar system Barycenter) that are measurable.

Usually the SSB is inside the Sun's volume, because the Sun makes up 99 per cent of the solar system's mass, but the position of the SSB can be outside the Sun as well, depending on where in an orbit Jupiter and Saturn are. For example, when Jupiter and Saturn are on the same side, the centre of mass obviously moves towards them, and thus the SSB can be located outside the Sun. The SSB over time is the image shown. It's a rookie mistake to make to assume that the orbits of the planets are fixed, and the Sun moves about in this way inside of those orbits. An error that you and I could make pretty easily, but for a PhD professor to make?

That's not all I'm booing though. How could peer review not catch such a simple error? And worse, if you read the scientific discussions about the paper, you find the author quickly devolving into name calling everyone else rather than accept her error. While Nature is investigating its own process to see if any lapses were made and how it can improve peer review, what's most troublesome for me is the invasion of politics and political leanings into the sacred field of science. You can read the rather frustrating discussion online (https://dgit.in/1908Boo2) but be forewarned, you will be dejected by the lack of civility shown by many, especially the author of the study. Is this where science and research is headed? Peer review being no different from pathetic social media arguments?